Devang Verma

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Education

Symbiosis Skills and Professional University Pune,India
Bachelor of Science in Data Science *July. 2024 – June 2027*

Vivek Techno School Jaipur, India
Middle School and High School *April. 2017 – May 2024*

Experience

Research Work Sep. 2025 – Feb. 2026
Spectral Drift–Diffusion Stability Criterion for Stochastic Dynamical Systems *India*

- Developed a spectral stability criterion for linear stochastic dynamical systems with multiplicative noise.
- Derived a closed-form eigenvalue condition for mean-square exponential stability.
- Proved that the proposed criterion strictly generalizes classical scalar contraction–diffusion stability bounds
- Constructed mathematical counterexamples demonstrating stability cases missed by traditional methods.

Research Work Dec. 2025 – Jan.2026
Bayesian Counterfactual Early-Warning System for ICU Deterioration *India*

- Proposed an uncertainty-aware temporal modeling framework for ICU deterioration prediction.
- Combined missingness-aware GRU-D modeling with Bayesian inference to capture uncertainty in clinical predictions.
- Developed a counterfactual risk simulation method for analyzing hypothetical early interventions

AI-ML Intern May 2025 – Aug.2025
AIT Global Inc *Pune, India*

- Developed Python-based tools to parse, transform, and standardize structured data formats (XML, DOCX) for downstream analytics systems.
- Designed a web interface for automated file ingestion and structured data extraction, enabling efficient processing of client data.
- Worked with XSLT and XML transformation pipelines to convert unstructured documents into machine-readable structured formats.
- Developed a prototype AI-powered document classification system using NLP techniques to categorize PDF content automatically and Collaborated with engineers on designing scalable data workflows for AI-driven applications

Projects

Engram | *Python, PyQt6, Ollama, Qdrant, Mistral 7B, Sentence Transformers, NetworkX* March. 2026 – Present

- Developing ENGRAM, a memory-centric artificial intelligence system conceptualized as a "second brain", which fundamentally redefines traditional chatbot paradigms by introducing persistent, structured memory layers, identity modeling, and deterministic control flows enabling the system to retain, retrieve, and reason over user-specific experiences across sessions rather than generating stateless, pattern-based responses.
- Preparing ENGRAM for a large-scale open-source launch as an early stage but foundational cognitive AI system, with a long-term vision of neuroadaptive systems and advancing toward real-world "second brain" infrastructure positioning it as a significant step beyond conventional AI assistants and as a potential building block for future personalized intelligence systems and human–AI cognitive augmentation

Blackwall | *Python, PyTorch, Graph Topologies , Multi-Agent RL , NetworkX* Jan. 2026 – Feb. 2026

- Built a five-phase system to detect, decode, and contain emergent language. four agents spontaneously developed a structured language with compositionality, semantic grounding, and covert steganographic channels
- Discovered a critical false-positive failure insider threat detector converged at 0.999 confidence on the wrong agent while the actual traitor scored 0.066 constituting emergent deceptive alignment without explicit programming; built a live 9-tab observatory dashboard streaming escape, suspicion, and decoder telemetry in real time.

Technical Skills

Languages:Python, C, SQL, JavaScript, HTML/CSS, R,
Frameworks: Ray RLlib, Hugging Face Transformers ,OpenAI Gym ,PyTorch, TensorFlow, Scikit-learn ,
AI/LLM Tools:Hugging Face Transformers, Ollama, LLaMA, Sentence Transformers, Embedding Models, Prompt Engineering